

Vishal L

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Career Objective

Motivated Computer Science Engineering undergraduate with hands-on experience in programming, web development, and database systems. Actively building real-world projects and participating in hackathons, seeking internship and skill-development opportunities in software engineering.

Education

Bachelor of Engineering – Computer Science and Engineering 2024 – 2028 (Expected)

Sri Sairam Engineering College, Chennai

Currently pursuing 3rd year (V Sem)

CGPA: 9.13

Higher Secondary Education (Class XII) 2024

Government Higher Secondary School, Marakkanam, Villupuram

Score: 88.83%

Secondary School Education (Class X) 2022

Government Higher Secondary School, Marakkanam, Villupuram

Score: 96.4%

Technical Skills

- **Programming Languages:** Python (Intermediate), C, Java
- **Web Development:** HTML5, CSS3, JavaScript, React, Next.js, FastAPI, Node.js
- **Databases:** PostgreSQL, MongoDB, MySQL
- **Version Control:** Git, GitHub, BitBucket
- **Core CS Fundamentals:** Data Structures (Basic), Object-Oriented Programming, DBMS, Operating Systems
- **Tools:** VS Code, Jupyter Notebook, Docker Desktop, PGAdmin, Prisma

Internships

Sri SaiRam Techno Incubator Foundation, Chennai 30 Days

Full Stack Development Intern

- Completed an intensive 30-day Full Stack Development internship covering modern web application development technologies, software engineering methodologies, and deployment practices.
- Gained hands-on experience in frontend and backend development using React, Next.js, FastAPI, PostgreSQL, REST APIs, Git, and GitHub.
- Designed, developed, and deployed the **Student Internship and Management System**, a web-based platform for managing internship lifecycles, attendance tracking, project allocation, and administrative operations.
- Implemented barcode-based attendance management, JWT authentication, RESTful APIs, and PostgreSQL database integration as part of the project development process.

- Worked with software development workflows including requirement analysis, database design, API development, testing, debugging, version control, and deployment.
- Developed practical understanding of system architecture, client-server communication, authentication mechanisms, database management systems, and scalable application design principles.

Projects

Ambulance-Despatch-RL-Model – Reinforcement Learning-Based Emergency Ambulance Dispatch Optimization System

GitHub Repository

- Developed a production-grade Reinforcement Learning environment for optimizing ambulance dispatch and allocation decisions in large-scale urban emergency response systems.
- Simulated realistic city-wide ambulance operations including emergency call generation, ambulance assignment, travel time estimation, hospital routing, and resource availability constraints.
- Designed the environment to model dynamic decision-making scenarios where agents learn optimal dispatch strategies to minimize emergency response times and improve service coverage.
- Implemented state representations, reward mechanisms, and action spaces suitable for training and evaluating Reinforcement Learning agents on ambulance dispatch optimization tasks.
- Built a scalable simulation framework capable of supporting experimentation with Deep Q-Networks (DQN), PPO, and other modern Reinforcement Learning algorithms.
- Evaluated system performance using operational metrics such as average response time, ambulance utilization, coverage efficiency, and dispatch effectiveness.
- Developed as part of the Scaler × Meta × HuggingFace × PyTorch OpenEnv Hackathon, focusing on AI-driven solutions for public healthcare and emergency management.
- **Tech Stack:** Python, Reinforcement Learning, OpenENV Framework

PromptForge – AI Prompt Engineering and Collaboration Platform

Project Repository

- Designed and developed a web platform for discovering, sharing, and managing high-quality AI prompts for real-world applications.
- Implemented features for creating, editing, forking, and bookmarking prompts, enabling collaborative prompt engineering among developers.
- Built a personalized user profile system with analytics including prompt activity graphs, reputation scoring, and contribution statistics.
- Integrated dual authentication methods (Google OAuth and email/password) with automatic avatar generation and editable user profiles.
- Developed interactive UI components including 3D hero sections, prompt visualization, and responsive dashboards for an engaging user experience.
- **Tech Stack:** React, Next.js, TypeScript, Tailwind CSS, Three.js, Supabase/PostgreSQL, Git, GitHub

Student Internship and Management System – Internship Lifecycle and Attendance Management Platform

Project Repository

- Designed and developed a centralized web-based platform to manage the complete lifecycle of student internships including student onboarding, project allocation, internship activation, and completion tracking.
- Built a barcode-based attendance system enabling batch scanning of student ID cards for fast and reliable internship attendance tracking with duplicate prevention and database integrity constraints.
- Developed internship lifecycle management with state transitions (pending, active, completed) and extension milestone tracking to monitor internship duration changes and maintain historical extension records.
- Engineered a structured RESTful backend API architecture supporting student management, project assignment, internship tracking, attendance recording, and administrative operations.
- Designed a scalable PostgreSQL database schema supporting relationships between students, projects, internships, attendance records, and extension histories.
- Implemented secure authentication and authorization using JWT tokens with protected API endpoints for administrative access control.
- **Tech Stack:** FastAPI, Python, PostgreSQL, Next.js, TypeScript, Tailwind CSS, AWS S3, ReportLab, JWT Authentication, Git, GitHub

Certifications

- Cisco – Python Essentials 1 & 2, Networking Devices and Basic Configuration
- IBM – SQL and Relational Databases 101, Python 101 for Data Science
- NPTEL – Joy of Computing Using Python, Programming in Java
- Udemy – Java OOP (Andril Partola, ITER Academy)
- La Wagon – Git and GitHub
- IIT Bombay Spoken Tutorial – Python and C
- Duolingo Hindi Score: 15 (Elementary Proficiency)

Hackathons and Competitions

- Finalist – OpenEnv Hackathon (Meta × PyTorch × Hugging Face × Scaler School of Technology) – India’s Largest Gen-AI Hackathon
- IEEE Xtreme 18.0 and 19.0 – Global Rank ~3000 Worldwide
- Finalist – IBM SkillsBuild Hackathon (24-hour national-level hackathon)
- Finalist – IdeaXchange Hackathon, VIT Chennai
- Finalist – Mumbai Hacks 2025
- Participant – Bharatiya Antariksh Hackathon 2025 (ISRO), Build for Billions Hackathon, Impact India Buildathon (Guvi × HCL), AI for Bharat (AWS × Hack2Skill), Fasthathon (SRM KTR), HackIndia Spark 6 (NIT Delhi), ArtPark CodeForge Hackathon (IISc Bangalore)